Meta-Agentic Program Synthesis v0

DEMO MASTER PRESENTATION

Sovereign orchestration • Verified synthesis • Governed execution

Agenda

A complete tour: architecture → orchestration → proof

Sovereign intelligence, compressed into a launch command

A non-technical owner can govern an autonomous architect that designs and verifies production-ready modules in minutes.

Operational promise

- Mission intent → verified artifacts (code + tests + dossiers).
- Owner supremacy: pause / thermostat / upgrades / treasury / compliance.
- Triangulated verification gates before deployment.

Invariant framing

$$\mathbf{Reward}(i) \propto e^{-E_i/(kT)}$$

Energy-optimal incentives: reward efficient work exponentially.

$$\nabla H(\mathbf{state}) = 0$$

Equilibrium: no utility leakage; stability under scale.

$$\Delta G \downarrow \;\Rightarrow\; \text{usable work} \uparrow$$

Free-energy gap shrinks as governance reduces waste.

A machine for verified synthesis at planetary scale

The demo shows a closed-loop system that discovers, proves, and ships solutions under strict owner governance.

Synthesis

Generate candidate pipelines and code modules via Quality-Diversity evolution.

Preserve a diverse frontier of high-quality solutions; avoid local minima.

Verification

Triangulate: deterministic replay, baseline dominance, adversarial jitter, constraint peer review.

Only consensus outputs ship.

Governance

Owner supremacy + council + sentinel grid.

Timelocks, contract limits, reward sanity, compliance scanning.

Mission → Archive → Proof → Artifacts → Learning

A closed loop: every run expands the capability frontier.

Meta-agentic orchestration with guardrails

A sovereign core coordinates agents, evolution, verification, and governance — with deterministic owner control.

Owner Control Plane

Launch · Pause · Thermostat
Upgrades · Treasury · Compliance

Sovereign Meta-Agent

Plans · decomposes missions
Coordinates specialized agents

Verification Fabric

Replay · baseline · jitter
Constraints · peer review

Specialized Agents

Vision · ledger · synthesis
Tools · test harnesses
Dashboards · dossiers

Evolutionary Forge (QD)

Explore diverse pipelines
Maintain global archive of elites

Governance + Sentinels

Council approvals
Timelocks · reward sanity
Contract bounds · anomaly watch

The 7-stage sovereign loop

A closed cycle that turns intent into deployed artifacts — then feeds learning back into the archive.

- **01** Launch — Manifest + safety defaults
- **02** Decompose — Spawn specialized agents
- **03** Evolve — QD search over pipelines
- **04** Verify — Triangulate + constraints
- **05** Package — Code + tests + dossier
- **06** Learn — Update archive + skill graph
- **07** Govern — Owner + council + sentinels

$$\mathbf{Archive}_{t+1} = \mathbf{Archive}_t \cup \mathbf{Elites}_t$$

Autopoietic compounding: elites become future priors.

Triangulated verification fabric

Multiple independent gates prove robustness before shipping.

Deterministic replay

Re-run candidate pipeline with zero drift.
Reproducibility is non-negotiable.

Baseline dominance

Must outperform known seeds or baselines.
No regression passes the gate.

Adversarial jitter

Perturb inputs and parameters.
Stability under noise is required.

Constraint + peer review

Bounds checks, safety constraints, and cross-candidate comparison.
Consensus determines shipment.

$$\mathbf{Ship} \iff \bigwedge_k \mathbf{Gate}_k(\mathbf{candidate})$$

Only when every gate agrees does the artifact become deployable.

Quality-Diversity evolutionary forge

Novelty is cultivated — not tolerated — while preserving elite performance.

Global QD archive

A 64-cell frontier of diverse elites.
Keeps multiple solution families alive.

Exploration ↔ exploitation

Thermostat tunes search pressure.
Find new basins without losing stability.

Elite synthesis

Hybridize and refine candidates.
Ship only verified, reproducible elites.

$$\max_{\pi} \ \mathbb{E}[\text{fitness}(\pi)] \quad \text{s.t.} \quad \text{diversity}(\Pi) \geq \tau$$

Optimization under a diversity constraint prevents brittle monocultures.

Three threads. One sovereign loop.

ARC · Ledger · Nova — cross-domain synthesis with full verification.

ARC — Sentinel Edge Lift

Vision / pattern discovery
Best score: 140.38
Accuracy: 100%
Energy target: 88 (Δ=0.00)

Ledger — Harmonics Sequencer

Economic stabilization
Accuracy: 100%
Energy target: 64 (Δ=0.00)
Governance checks: pass

Nova — Weave Optimiser

Code synthesis / on-chain blueprint
Score: 136.05
Accuracy: 100%
Constraints: aligned

$$\textbf{Accuracy} = 1.00 \wedge \bar{\Delta} = 0.00 \wedge \textbf{Novelty} \approx 0.78$$

Perfect correctness with thermodynamic alignment and high novelty signal.

ARC Sentinel Edge Lift

Computer vision: detect latent edges; amplify faint structure; feed skill graph.

Objective

Edge extraction under low signal.
Deliver a robust transformation that generalizes.

Result

Best candidate score: 140.38
Accuracy: 100%
Energy target: 88 (Δ=0.00)

Why it matters

Demonstrates perception pipelines discovered and proven under the same sovereign loop.

Verification

- Replay: reproducible output
- Dominance: beats seed
- Jitter: stable under perturbations
- Constraints: within energy envelope

$$\Delta = |E - 88| \to 0$$

Thermodynamic alignment: exact energy target reached.

Ledger Harmonics Sequencer

Stabilize asynchronous cash-flow deltas against a thermodynamic ledger; emit rebalanced incentive curves.

Objective

Absorb shocks; maintain stable incentives under governance constraints.

Result

Accuracy: 100%
Energy target: 64 (Δ=0.00)
Governance checks: pass

Mechanism sketch

$$r_i \propto e^{-E_i/(kT)}$$

Efficient validators receive higher reward share.

$$\textbf{Stability}: \Delta G \downarrow$$

Reduce inefficiency to increase usable work.

$$\Delta = |E - 64| \to 0$$

Exact energy budget alignment for the run.

Nova Weave Sequence Optimiser

Elevate alpha sequences into deterministic blueprints that compile into deployable payloads with zero manual edits.

Objective

Transform high-level sketches into verified deployable code modules.

Result

Score: 136.05
Accuracy: 100%
Constraints: aligned (gas/size)

Deployment integrity

- CI gates green
- Reproducible builds
- Hash + dossier for audit
- Upgradeable only via governance

$$\mathbf{Ship} \Rightarrow \mathbf{Audit}(\text{hash},\ \text{tests})$$

Artifacts are verifiable by construction.

Performance & reliability

Correctness, novelty, and efficiency — jointly optimized.

100% Accuracy	**100%** Coverage	**0.00** Δ energy deviation
77.76% Novelty signal (avg)	**73.33** Energy envelope (avg)	**140.38** Global best score

Proof & Efficiency & Novelty → Ship

The system optimizes multi-objective performance without sacrificing safety.

Owner supremacy by design

All high-impact operations are directly controllable and auditable.

Emergency pause

Hard stop for all agents and execution.

Thermostat

Tune exploration/exploitation and resource envelope.

Upgrades

Module updates via governance + timelock.

Treasury & rewards

Bounded incentives; reward sanity checks.

Compliance scan

Continuous monitoring and dossier generation.

$$\mathrm{Owner}(t) \succ \mathrm{Agent}(t)$$

The agent cannot override owner commands or policy.

Council oversight + sentinel grid

Institutional guardrails that scale with capability.

Governance council

- Multisig approval for critical changes
- Timelocked upgrades
- Policy bounds enforced on-chain

Sentinel grid

- sentinel.quantum
- sentinel.reward-engine
- sentinel.timelock
- sentinel.contract-size

Effect

Bounded power

High capability remains governable:
constraints, auditability, and layered checks are always on.

$$\textbf{Action} \in \mathcal{P}\ \text{(policy set)}$$

Only policy-allowed actions can execute.

Thermodynamic incentives

Energy accounting turns coordination into a self-stabilizing engine.

Energy → reward

Efficient work is favored exponentially; waste is suppressed by design.

$$r_i \propto e^{-E_i/(kT)}$$

Maxwell–Boltzmann-weighted reward share.

$$\mathbf{Thermostat:}\ T\uparrow \Rightarrow \text{explore}$$

Tune temperature to control exploration pressure.

Implication

Minimum-waste attractor

At scale, the system gravitates toward low-energy, high-utility regimes — like a physical ensemble approaching equilibrium.

Free-energy gap as a coordination lens

Reduce inefficiency; increase usable work.

Definition

Let ΔG represent the gap between current output and theoretical maximum utility under perfect efficiency.

Verification + governance reduce ΔG by eliminating waste.

Mechanism

Disputes, drift, and idle work raise ΔG.
Penalties + bounds + incentives drive ΔG downward.

$$\Delta G = \mathrm{waste}(t) \quad \Rightarrow \quad \Delta G \downarrow \ \text{over time}$$

A system that can’t leak utility becomes a compounding engine.

Hamiltonian stability + repeated-game alignment

An equilibrium where cheating is dominated by cooperation.

Hamiltonian view

$$H(\text{state}) = K + V$$

K: resources; V: future utility.

$$\nabla H = 0$$

Balanced equilibrium: no net utility leakage.

Repeated-game alignment

$$\delta \uparrow \Rightarrow \text{shadow of the future}$$

Future rewards dominate one-shot cheating.

- Stake + slashing
- Delayed rewards
- Reputation & access control
- Timelocked upgrades

Toward Type-II cognition

Capture and focus civilization-scale problem-solving energy — safely, for all sentient beings.

Analogy

A Type-II civilization harnesses star-scale energy.
Here, the engine harnesses star-scale coordination: many agents, one governed mission.

What scales

- Parallel synthesis + verification
- Auditability and policy boundaries
- Energy-optimal incentives
- Autopoietic learning loops

This is how capability becomes infrastructure: governed, repeatable, compounding.

Utility coordination layer

A pure utility unit for mission execution, governance, and incentives.

Execution

Fuel jobs, fees, and bounded resource allocation for missions.

Governance

Weight protocol decisions: thermostat, upgrades, council parameters.

Incentives

Reward verified work and honest validation under energy accounting.

$$\textbf{Utility} \Rightarrow \textbf{Coordination} \Rightarrow \textbf{Compounding capability}$$

Utility is the actuator; governance is the boundary.

Deploy a sovereign mission

A demonstration of governed power: synthesis, verification, and compounding capability.

What this demo proves

- A sovereign meta-agent can autonomously produce deployable artifacts under strict governance.
- Quality-Diversity evolution discovers novel high-performance solutions.
- Triangulated verification makes outputs safe to ship.
- The archive compounds: each mission expands the capability frontier.

Gateway

Run missions · audit dossiers · govern upgrades · coordinate utility ($AGIALPHA)

montreal.ai · AGI Jobs v0 · Meta-Agentic Program Synthesis v0

Meta-Agentic Program Synthesis v0

From intent to deployed reality — governed, verified, scalable.

Primary sources:

github.com/MontrealAI/AGIJobsv0/tree/main/demo/Meta-Agentic-Program-Synthesis-v0